

AutoNerve

A unified AI nervous system connecting the supply chain to the shop floor.

Design & Build Document

ET AutoTech Hackathon 2026 | Theme 1: Resilient Automotive Supply Chains & Smart Manufacturing
Prototype-round build plan | deliverables: PPTX + demo video
Author: Mrinal | Status: design locked, build in progress

1. What AutoNerve is

AutoNerve is a single AI system that reasons across two signal worlds at once: external supply signals (news, geopolitical events, commodity prices) and internal shop-floor signals (quality, process capability, energy). Both feed one shared brain that drives two surfaces of action - a **Supply Cortex** for procurement and sourcing, and a **Plant Cortex** for the shop floor.

The differentiator is the **closed loop between the two**: a sourcing decision knows its quality consequence, and a quality signal on the floor changes the sourcing decision. Most teams build one half well; AutoNerve treats a procurement decision and its manufacturing consequence as the same reasoning problem.

2. Core idea: the LLM decides, deterministic tools compute

The intelligence sits in **how an LLM makes decisions** - but with a strict boundary that keeps it credible:

- **The LLM decides.** It interprets a messy real-world signal, works out the BOM consequences, chooses which tools to run, weighs the trade-offs they surface, and commits to a recommendation it can justify - senior-planner judgment, automated.
- **Deterministic tools compute.** Graph propagation, MRP explosion, and the optimizer produce every number (risk scores, allocation percentages, exposure), so the LLM never fabricates a figure. This matters because a quantized local model freelancing the math is the one thing a domain judge distrusts on sight. Grounded decisions win; freelanced ones lose. The novelty is not the optimizer (operations research is decades old) - it is that an LLM runs the whole **sense -> reason -> decide -> act** loop autonomously and explainably in a domain that is traditionally rules-engines and spreadsheets.

3. Architecture: three layers

- **LLM reasoning layer** - LangGraph orchestration; interprets signals, sequences tools, synthesises decisions with rationale (local Mistral 7B).
- **Deterministic tool layer** - risk propagation, MRP/demand explosion, the cost-vs-procurement optimizer (MILP), scoring formulas. These produce defensible numbers.
- **Central knowledge graph (the BOM)** - the grounding substrate every layer reads and writes. Every decision anchors to nodes and edges the model cannot invent. The graph grounds; the LLM decides. That is the architecture-clarity story in one diagram.

4. The central knowledge graph (BOM)

A supply-chain knowledge graph - typed entities and typed relationships traversed and reasoned over - implemented on Postgres (relations as columns / foreign keys, multi-hop via recursive CTEs). It is graph-semantic, not an RDF triple store, and reasoning is multi-hop traversal rather than formal inference.

Entities & relationships

- **Entities:** products (5 main assemblies), sub-assemblies, components, raw materials, commodities, suppliers, regions.
- **Relationships:** contains, made-of, priced-by, supplied-by, located-in, substitutable-by, used-in.
- **Three traversal directions:** down (BOM explosion via qty_per_unit), up (where-used roll-up to finished goods), sideways (alternates / substitution).

Five main assemblies (permuted parts pool)

EV Sedan Performance, EV Sedan Economy, ICE Sedan, EV SUV, ICE MPV. Shared parts (chip, copper, brakes, fasteners) cascade across all five; rare-earth parts hit only the NdFeB EVs; the Economy line is already insulated via a ferrite substitute. Different events therefore light up different subsets - the property that makes propagation feel alive instead of everything turning red at once.

Key BOM columns

Group	Columns
Identity / structure	part_id, part_name, part_type, parent_part_id, qty_per_unit, bom_level
Material	material, material_grade, linked_commodity
Supplier & geo	supplier_id, supplier_name, supplier_tier, country, lat, lon
Economics	unit_price_inr, quality_adj_cost_inr, lead_time_days, capacity_per_week
Quality / risk	incoming_defect_rate, china_dependency_pct, risk_score, iatf_certified
Relationships	alternate_part_ids, used_in_assemblies, main_assembly
Flags	is_primary, qualification_effort, feature_tier_impact

Quality enters the economics as a quality-adjusted cost ($\text{price} / (1 - \text{defect rate})$), so the optimizer minimises cost *per good unit*, not sticker price - the mechanism behind the risk-adjusted TCO figures.

5. Data assets generated

- **bom.csv** - 31 rows, 29 columns. The knowledge graph: 5 products + sourcing options + alternates, deck-consistent numbers.
- **demand_series.csv** - 26 weeks history + 12 weeks forecast with uncertainty bands. Products forecasted directly; parts derived by BOM explosion (dependent demand, not forecasted independently).
- **commodity_prices.csv** - copper, rare-earth, steel, silicon, aluminium price history + forecast with bands.

All synthetic, for the demo. Framing on the deck: "demonstrated on synthetic history; ingests actuals from ERP/MES in production."

6. The bidirectional closed loop

- **Supply -> Plant (substitution-aware quality):** a sourcing switch is checked for its Cpk impact before it is recommended.
 - **Plant -> Supply (quality-aware sourcing):** defects detected on the floor raise a supplier's incoming defect rate, downgrade its quality score, and trigger the optimizer to re-prioritise alternates against open POs.
- Both directions feed the **same optimizer re-solve**, so making the return loop live is nearly free once the forward loop exists. This is the moat - no single-half team can show it.

7. Decision flow - worked example

The neodymium scenario, traced end to end with the actual data (this is the demo climax and the strongest single deck slide):

- **Event:** China announces export controls on neodymium magnet grades.
- **KG link:** matches material -> raw node RM-ND; used_in_assemblies flags NM-005 -> ROT-118 -> EVSED-P + EVSUV exposed.
- **Propagation:** composite risk 0.82 on the NdFeB supplier; SHAP-style attribution from contributing columns.
- **Demand explosion:** ~6,440 magnets/week at risk across the exposed products.
- **Optimizer:** cheapest mix that zeros China-dependency = switch 40% to Sundram, +6% cost, Cpk impact <= 0.03.
- **Act:** PO recommendation pushed; dashboard re-solves live.

8. Capability coverage (slides 7-22)

Status reflects design + data readiness; live wiring is the remaining build. **Strong** = designed, data in the BOM, part of the live spine. **Partial** = a real piece + seeded rest. **Gap** = not addressed.

#	Capability	Status
7	Risk & Bottleneck Detection	Strong
8	Predictive Planning	Partial
9	Alternate Sourcing	Strong
10	Material Substitution	Strong
11	Commodity Intelligence	Partial
12	Production Data Analysis	Partial
13	Process Capability (Cp/Cpk)	Partial
14	Low-Cost Quality Automation	Partial
15	Inventory Forecasting	Strong
16	Energy Prediction	Gap
17	Supplier Analytics	Strong
18	ERP Decision Engine	Strong
19	Meeting & Action Extraction	Partial
20	Computer Vision Defect Detection	Partial
21	Operator AI Guidance	Partial
22	Real-Time Quality Monitoring	Partial

The pattern is intentional: the Supply Cortex (7-11) plus the decision engine, supplier analytics and inventory are the strong live spine - the geopolitical-risk to de-risking-action thread the rubric emphasises. The Plant-quality block is honest seeded breadth. The one real gap is Energy (16).

9. Build strategy

"Everything integrated" means one coherent system - one data model, one shared brain, one UI - not sixteen trained models. Every panel hangs off the same backbone; a judge cannot tell a seeded panel from a live one when they read the same API. What they can see is whether it is one nervous system or six scripts.

Rubric alignment

60 of 100 points (Problem Insight 20, Innovation 20, Architecture Clarity 20) live in the deck and how it is explained. Feasibility (15) is where the live spine earns its keep; Impact & Scalability (15) wants the business model; Presentation/Demo clarity (10) is the video. The build exists to be filmed - it is judged through a PPTX and a video, with no code submission - so build the one worked-example thread, not the whole product.

10. Technology stack

Prototype (what runs)

FastAPI + Postgres (with pgvector) + Ollama + Next.js. Local models: Mistral 7B (reasoning/extraction), moondream or qwen2.5vl (vision), BGE-M3 (embeddings). Fully on-prem, no data leaves the plant, zero per-token cost, edge-deployable - itself a feasibility and impact argument.

Target (on the deck, labelled)

Kafka ingestion, TimescaleDB sensor series, MinIO image archive, Redis cache, Celery async jobs; R-GCN risk model, XGBoost/TFT forecasting, NHITS/DeepAR, YOLOv8/PatchCore vision. Each defensible in one line; trim anything that cannot be justified to avoid a buzzword-stuffing read.

11. Build plan / timeline (deadline Thu Jun 4)

- **Mon - the spine:** the one live thread end to end - event -> KG link -> propagation -> MRP exposure -> optimizer -> recommendation - plus a minimal Control Room + worked-example view to film. Run off CSVs in memory if Postgres is slow.
 - **Tue:** 3D globe (hero shot), the remaining hero screens, the cheap live adds (meeting, operator, vision) if time.
 - **Wed:** layered deck - clean main arc + appendix; the three high-leverage slides (3-layer architecture, KG schema, worked-example trace); the business-impact / scalability model.
 - **Thu:** record the 2-4 min video around the optimizer re-solve climax; submit PPTX + video.
- Depth is not density: a clean main arc up front, everything micro in the appendix, one ruthlessly-worked example as the proof.

12. Honest framing (claim vs do not claim)

- Knowledge graph on Postgres - **yes**; RDF/OWL triple store - **no**.
- LLM-orchestrated, grounded decisions - **yes**; LLM computing the numbers - **no**.
- Representative BOM / synthetic history, ingests real data in production - **yes**; ground-truth discovered BOM - **no**.
- Scenario simulator / what-if disruption injector - **yes**; "we generate news" - **no**.
- Deterministic propagation + traversal - **yes**; trained GNN/XGBoost running live - **no** (target).

13. Decisions

Locked

- One backbone, fidelity varies (one system, not 16 models).
- Local Ollama stack; deterministic propagation; GNN as target.
- Heavy infra (Kafka/Redis/etc.) = target only; prototype = FastAPI + Postgres + Ollama + Next.js.
- Central knowledge-graph BOM on Postgres; relationships as columns.
- 5 main assemblies permuting a shared parts pool.
- 3D = global map (globe), hero video shot + 2D for drill-down.
- AI roles: build-time seed generation, live news linkage, deterministic propagation.

Open

- Live-sync: SSE (leaning yes) vs polling.
- Seed accuracy: grounded hero families + AI filler vs fully AI-generated.
- Optimizer objective: cost-min with a geopolitical-dependency constraint vs weighted blend.
- Energy (16): seeded panel vs drop from demo.